

LECTURE 14

MANAGING PEOPLE IN SOFTWARE ENVIRONMENT

Knowledge Area	Process				
	Initiating	Planning	Executing	Monitoring & Control	Closing
Human Resource Management		Develop HR Plan	Acquire Project Team Develop Project Team Manage Project Team		

Introduction to Human Resource Management Process

In the Human Resource Management process, you plan out exactly which resources you'll need, what their roles and responsibilities are, and how you'll train your team and make sure they stay motivated.

Human Resource Management Processes

- Develop Human Resource Plan
- Acquire Project Team
- Develop Project Team
- Manage Project Team

Develop Human Resource Plan Process:

The process of identifying and documenting project roles, responsibilities and required skills, reporting relationships and creating a staffing management plan.

Tools & Techniques:

- All team members have a clear understanding of their roles and responsibility.
- Hierarchical e.g. Organizational Breakdown Structure (OBS) as shown in Figure 1.
- Matrix e.g. Responsibility Assignment Matrix (RAM) e.g. RACI (responsible, accountable, consult, inform) as shown in Figure 2 & 3.

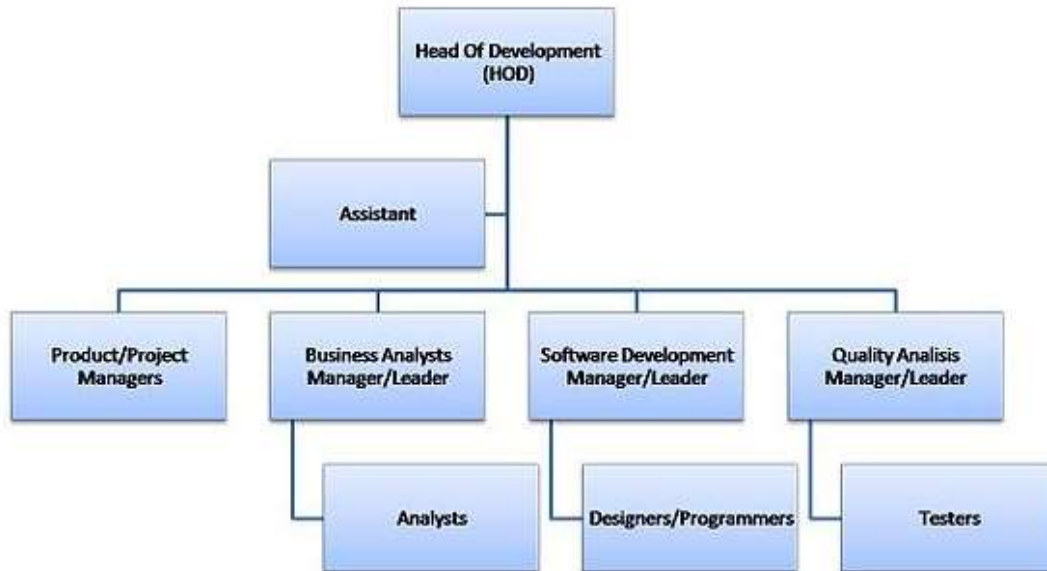


Figure 1: Organizational Breakdown Structure (OBS)

RACI matrix roles and responsibilities

The RACI model brings structure and clarity to describing the roles that stakeholders play within a project. The RACI matrix clarifies responsibilities and ensures that everything the project needs done is assigned someone to do it.

RACI Chart

	Project Manager	Strategist	Designer	Front End Developer	Back end Developer
Design sitemap	C	R	A	I	I
Design wireframes	C	A	R	I	I
Create style guide	A	C	R	C	I
Code templates	A	I	C	R	C

R = Responsible A = Accountable C = Consulted I = Informed

Figure 2: RACI Chart

Step	Project Initiation	Project Executive	Project Manager	Business Analyst	Technical Architect	Application Developers
1	Task 1	C	A/R	C	I	I
2	Task 2	A	I	R	C	I
3	Task 3	A	I	R	C	I
4	Task 4	C	A	I	R	I

CID/IDG

Figure 3: RACI Chart

The four roles that stakeholders might play in any project include the following:

Responsible: People or stakeholders who do the work. They must complete the task or objective or make the decision. Several people can be jointly Responsible.

Accountable: Person or stakeholder who is the "owner" of the work. He or she must sign off or approve when the task, objective or decision is complete. This person must make sure that responsibilities are assigned in the matrix for all related activities.

Consulted: People or stakeholders who need to give input before the work can be done and signed-off on. These people are "in the loop" and active participants.

Informed: People or stakeholders who need to be kept "in the picture." They need updates on progress or decisions, but they do not need to be formally consulted, nor do they contribute directly to the task or decision.

Acquire Project Team

The process of confirming human resource availability and obtain the team necessary to complete project assignments. Read as "Acquire final project team."

Tools & Techniques:

- Pre-assignment
 - Resources who are assigned in advance
- Negotiation
 - For gaining resources within the organization or external vendors, suppliers, contractors, etc (in contract situation)

- Acquisition
 - Acquiring/hiring from outside resources (outsource)
- Virtual teams
 - Think the possibilities of having group of people even little or no time spent to meet face to face.

Develop Project Team Process:

The process of improving the competencies, team interaction and the overall team environment to enhance project performance.

Tools & Techniques:

- Interpersonal skills (soft skills)
- Training
 - Can be formal (classroom. online) or non-formal (on-job training, mentoring, coaching)
- Ground rules
 - Guidelines that establish clear expectation regarding acceptable behavior by teams
 - Discussion to create it by all team members
- Co-location
 - Placing many or all the most active team members in the same physical location
 - Can be temporary for strategy to enhance communication & build sense of community
- Recognition & reward
 - It will only be effective if it is satisfies/valued by individual.
 - Plans concerning how to do it are developed during Develop Human Resource Plan.

Team Building Activities Process:

There is a process for a team to get from a group of strangers to a group that creates something good together, and that is what the stages of team development are all about.

Tools & Techniques:

Tuckman's stage of team formation and development:

- FORMING
 - The team meets and learns about the project and what their roles and responsibilities.
- STORMING

Address the project work. Technical decisions and the project management approach. Conflict/disagreement may occur.

- NORMING
Work together and adjust work habits and behavior that support the team.
- PERFORMING
Being a well-organized unit
- ADJOURNING
Team completes the work and move on from the project.

Manage Project Team Process:

The process of tracking team member performance, providing feedback, resolving issues, and managing changes to optimize project performance.

Tools & Techniques:

Conflict Management:

- Conflicts **can be beneficial** (an opportunities for improvement)
- Conflicts are **an inevitable consequence** of organizational interactions.

Conflicts in the team are caused due to the following reasons in decreasing order of occurrences.

- Schedules
- Project priorities
- Resources
- Technical opinions

The most common cause of conflicts in projects is issues related to schedules (not personality differences).

Conflict Resolution Techniques

The PMBOK Guide sixth edition lists five conflict resolutions techniques as shown in Figure 4:

1. Withdraw/Avoid
2. Smooth/Accommodate
3. Compromise/Reconcile
4. Force/Direct
5. Collaborate/Problem Solve

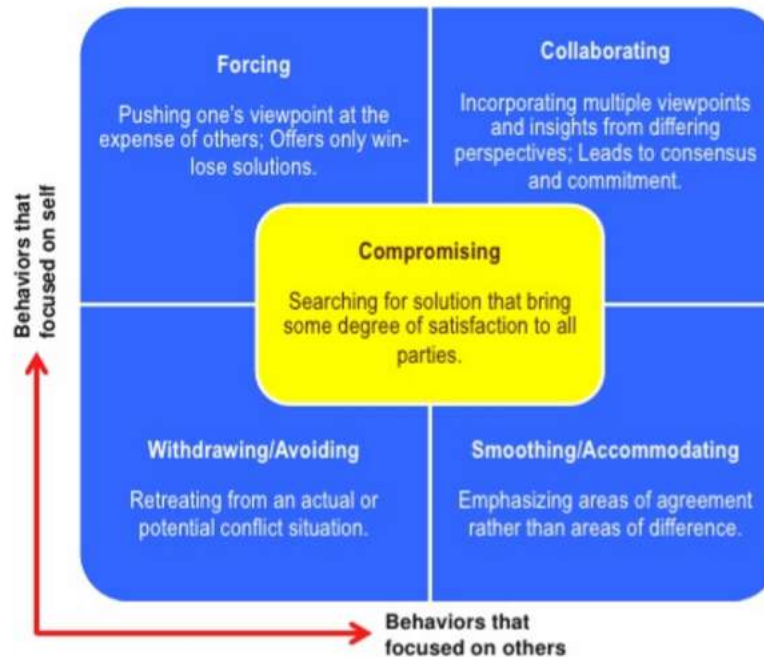


Figure 4: Five conflict resolutions techniques

Withdraw or Avoid

In this conflict resolution technique, you avoid the conflict or retreat and allow it to resolve itself. This technique is beneficial when stakes are low and it is likely the conflict will disappear on its own.

This technique can be used in the following cases:

- Individuals involved in the conflict are not major stakeholders.
- The issue does not require a great time investment.
- An intense argument has already happened, and the individuals need time to cool off.
- If you do not have enough information to pursue other techniques.

Advantages

This technique saves time that you can invest in other productive activities. It is a good approach to apply to low-level conflicts and gives you enough time to prepare if the conflict re-emerges. .

Disadvantages

Withdrawing or avoiding a conflict may weaken your position as a project manager because parties may assume you have an unfair bias. Team members may think you are lacking skills and are not authoritative.

Smooth or Accommodate

This technique deals with finding areas of agreement and tries to smooth the situation and circumvents tough discussions.

In smoothing, you give more consideration to one party than the other. Here you try to downplay the seriousness of the situation and behave as if the problem never existed.

This technique is useful in the following cases:

- There is not enough time to deal with the conflict.
- When only a temporary solution to the problem.
- If the conflict is, minor and involves less-influential stakeholders.

Advantages

This technique does not require much effort. You can focus on essential issues while ignoring unimportant arguments. The situation can be handled simply while, bringing harmony, creating goodwill, and providing enough time to find a permanent solution.

Disadvantages

If you fail to bring a balanced approach to smoothing, one party may take advantage of the situation since you are giving them more consideration. Members of the party not being accommodated to may question your authority or stop reporting conflicts.

This technique is not recommended as it often weakens the project manager's authority.

Compromise or Reconcile

This is where you take suggestions from both sides and try to partially satisfy both parties. This technique is useful when the stakeholders involved in the conflict hold equal power.

You may use this technique in the following cases:

- All parties involved in the conflict need to win.
- When you, the project manager, have an equal relationship with both parties.
- Collaborative and forcing techniques have not worked.
- When you need a temporary solution to move forward quickly.

Advantages

This technique brings quick results, lowers stress, and keeps all parties placated until you can find a permanent solution. You can solve a conflict and gain enough time to find a better solution.

Disadvantages

This technique does not generate trust in the long run; all parties remain unsatisfied, and the conflict could resurface at any time. Moral is not being built. You may be required to interfere sooner to make sure that all parties abide by the agreement.

Force or Direct

Here you agree with one party's viewpoint and enforce their wishes. This is a win-lose situation and risks demoralizing the team.

This conflict resolution technique can be used in the following cases:

- When you need a quick solution.
- When you know that one party is right and you do not have time to investigate.
- When the stakeholders on the losing side of the conflict are not very important.
- If the relationship with those stakeholders is not essential.

Advantages

This technique provides a quick solution to the problem. It requires almost no effort from the project manager, and it may help establish the project manager's authority.

Disadvantages

Using this technique may cause a negative impression of you. You may lose opportunities gained from the opposing party's viewpoint. You cannot apply this technique with high power stakeholders. Sometimes it may backfire and worsen the conflict.

Collaborate or Problem Solve

In this technique, you discuss the issue with all parties to agree on a solution, while considering multiple viewpoints.

You may use this technique in the following cases:

- When you want to incorporate multiple views.
- If the people involved in the conflict are very influential.
- When a consensus is required.
- If you want to distribute responsibility equally to all parties.

Advantages

This is a real problem-solving technique and gives a final solution to the conflict. It brings consensus, commitment, and shared responsibility for the outcome. This technique creates a win-win situation. It builds your confidence in the team, earns you respect and establishes your authority.

Disadvantages

You cannot use this technique when you need a quick solution because it takes time and effort. You cannot utilize this technique with all conflicts; it is generally used for conflicts which may severely affect your project.

Project Manager Power:

A Project Manager may yield authority over the project team in one of the following ways

1. **Formal** (Legitimate) - Power due to Project Managers position
2. **Reward** — Power stems from giving rewards.
3. **Penalty** (Coercive) — Power due to afraid of the power the Project Manager holds.
4. **Expert** (Technical) — Comes from being technical or project management expert.
5. **Referent** — Power of charisma and fame. Make another person liking/respecting the Project Manager.

Points to remember:

- ✓ The best forms of power: EXPERT and REWARD.
- ✓ Earned on your own: EXPERT
- ✓ The worst choice: PENALTY
- ✓ Derived from position in the company: FORMAL, REWARD and PENALTY.

Management & Leadership Style

- Autocratic
 - Top down approach. The manager has power to do whatever she/he wants.
 - Sometime appropriate when decisions must be made for emergency situation or time pressure.
- Democratic/Participative

- Encouraging team participation in the decision making process
- Best used for people whose behavior fit with theory Y
- Laissez-faire • a French term means 'leave alone'
- The manager is not directly involved in the work of the team.
- Effective for highly skilled team

Important Terms

Halo Effect

The assumption that because the person is good at technical, he will be good as a project manager.

Arbitration

A method to resolve conflict. Neutral parties hear and resolve a dispute.

Expectancy Theory - H. Vroom

This is a motivation factor. People put in more efforts because they expect to be rewarded for their efforts.

Perquisites (Perks)

Some employees receive special rewards e.g. parking spaces, corner offices, executive dining.

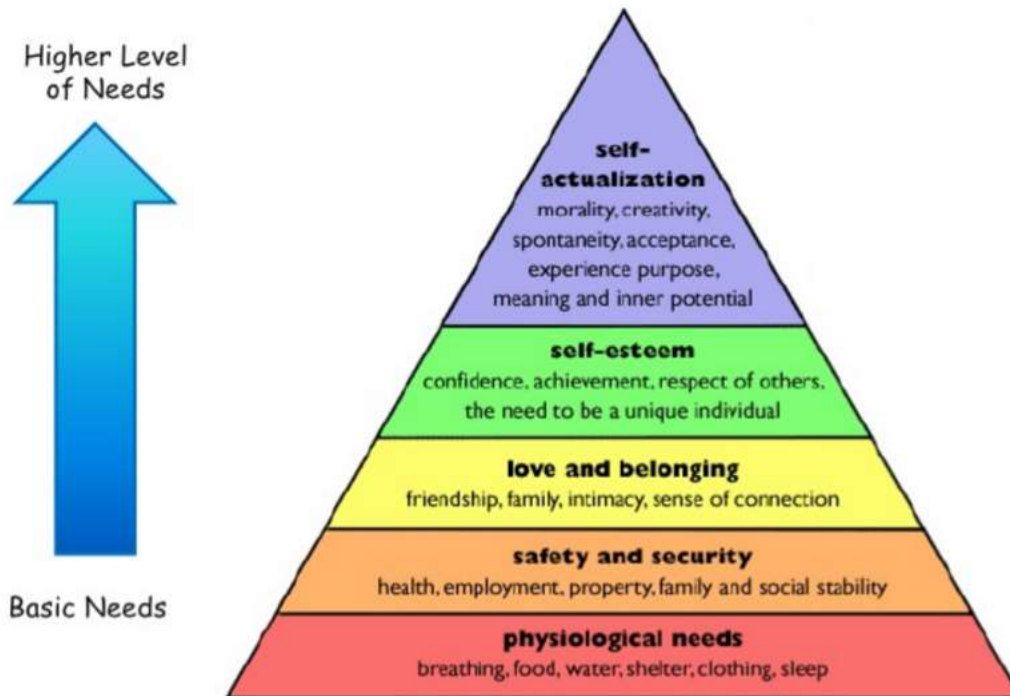
Fringe Benefits

Standard benefits formally given to all employees, such as insurance, education benefits and profit benefits.

Project Manager Interpersonal Skills

- Leadership
- Team building
- Motivation
- Communicating
- Influencing
- Decision Making
- Political and cultural awareness
- Negotiation

Motivation Theory: Maslow's Hierarchy of Needs



Motivation Theory: McGregor's X & Y Theory

- Theory X
 - People tends to be negative, passive e.g. incapable, avoid responsibility, need to be watched
 - Extrinsic Motivation
- Theory Y
 - People tends to be positive e.g. want to achieve, willing to work without supervision, can direct their own effort
 - Intrinsic Motivation

Motivation Theory: Acquired Needs Theory

- David McClelland's Theory

People are motivated by one of the three needs

Needs	Behavioral Style
Achievement (N-Ach)	- These people should be given projects that are challenging but are reachable - They like recognition
Affiliation (N-Affil)	- These people work best when cooperating with others - They seek approval rather than recognition
Power (N-Pow)	- People whose need for power is socially oriented, should be allowed to manager others - These people like to organize and influence others

Motivation Theory: Two Factors Theory

- Herzberg's Theory
 - Job dissatisfaction due to lack of hygiene factors
 - Job satisfaction due to motivation factors

